

Environmental Heat Monitoring Dashboard Report

Business Problem Analysis

Problem Statement

Environmental heat conditions can increase the chances of overheating and fire-related risks in industries, offices, storage facilities, and manufacturing environments. Manual monitoring of temperature and humidity is difficult and time-consuming.

Organizations need a smart monitoring system that:

- Tracks temperature and humidity levels
- Calculates effective heat values
- Identifies burn-risk conditions
- Provides visual reports for decision-making

Introduction

The Environmental Heat Monitoring Dashboard is a data analytics project developed to monitor and analyze environmental heat conditions using temperature, humidity, and effective heat values. The project helps in identifying risky heat conditions that may cause paper burn or overheating situations.

The dashboard provides real-time visual analysis of heat trends, humidity groups, and burn conditions using interactive Power BI visualizations.

Tools and Technologies Used

Tool/Technology	Purpose
Power BI	Dashboard creation and data visualization
Microsoft Excel	Data preparation and cleaning
DAX (Data Analysis Expressions)	Calculations and measures
Power Query	Data transformation

Dashboard Features

KPI Cards

The dashboard contains important KPI indicators:

- Average Temperature
- Average Humidity
- Average Effective Heat
- Burn Percentage

Charts and Visualizations

1. Scatter Plot

Effective Heat & Lens Factor by Burn Paper

- Shows relationship between temperature and effective heat
- Identifies burn-risk conditions

2. Line and Bar Chart

Daily Trend of Effective Heat & Temperature

- Displays heat trends across dates
- Helps identify peak heat days

3. Donut Chart

Burn Paper Distribution

- Shows percentage of burn and non-burn conditions

4. Bar Chart

Effective Heat by Day Name

- Analyzes heat values for each day of the week

5. Table Visualization

- Displays complete environmental monitoring records

Business Recommendations

- Install real-time environmental monitoring systems
- Set automated heat-risk alerts
- Improve ventilation and cooling systems
- Maintain optimal humidity levels
- Perform regular equipment maintenance
- Use predictive analytics for heat forecasting
- Train employees on heat safety measures

